

1 Probabilistic Models

1.1 Three-Parameters Weibull Distribution

This is a modification of the two-parameter Weibull model by adding a threshold δ in σ_f in eq (). The CDF of this model is given by:

$$f_w = 1 - \exp\left(-\left(\frac{\sigma_f - \delta}{\sigma_m}\right)^m\right) \quad (1)$$

PDF of this model can be expressed as:

$$F_w = \left(\frac{m}{\sigma_m}\right) \left(\frac{\sigma_f - \delta}{\sigma_m}\right)^{m-1} \exp\left(-\left(\frac{\sigma_f - \delta}{\sigma_m}\right)^m\right) \quad (2)$$

Eq.1 can be rearranged as

$$\sigma_f = \exp\left(\ln(\sigma_m) + \frac{1}{m} \ln\left(\ln\left(\frac{1}{1-f_w}\right)\right)\right) + \delta \quad (3)$$

Similar to two-parameters Weibull model, using Arrhenius Eq, Eq 3 can be represented as:

$$\sigma_f = \exp\left(\left\{U_t + \frac{W_t}{T}\right\} + \frac{1}{m} \ln\left(\ln\left(\frac{1}{1-f_w}\right)\right)\right) + \delta \quad (4)$$

Fatigue life prediction model can be presented as

$$\sigma_f = \exp\left(\left\{U_t + \frac{W_t}{T} + V \ln(\epsilon)\right\} + \frac{1}{m} \ln\left(\ln\left(\frac{1}{1-f_w}\right)\right)\right) + \delta \quad (5)$$

1.2 Weibull Distribution With Power Law

The CDF of this model is given by:

$$f_w = 1 - \exp\left(-\left(\frac{\sigma_f}{\sigma_m}\right)^m\right) \quad (6)$$

PDF of this model can be expressed as:

$$F_w = \left(\frac{m}{\sigma_m}\right) \left(\frac{\sigma_f}{\sigma_m}\right)^{m-1} \exp\left(-\left(\frac{\sigma_f}{\sigma_m}\right)^m\right) \quad (7)$$

Eq.6 can be rearranged as

$$\sigma_f = \exp\left(\ln(\sigma_m) + \frac{1}{m} \ln\left(\ln\left(\frac{1}{1-f_w}\right)\right)\right) \quad (8)$$

using Power Law Eq, Eq 8 can be represented as:

$$\sigma_f = \exp\left(\left\{U_t + W_t \ln(T)\right\} + \frac{1}{m} \ln\left(\ln\left(\frac{1}{1-f_w}\right)\right)\right) \quad (9)$$

Fatigue life prediction model can be presented as

$$\sigma_f = \exp\left(\left\{U_t + W_t \ln(T) + V \ln(\epsilon)\right\} + \frac{1}{m} \ln\left(\ln\left(\frac{1}{1-f_w}\right)\right)\right) \quad (10)$$

1.3 Three-Parameters Lognormal Distribution

This is a modification of the two-parameter lognormal model by adding a threshold γ in σ_f in eq (1). The PDF of this model is given by:

$$F_{ln} = \frac{1}{(\sigma_f - \gamma)\sigma_{sl}\sqrt{2\pi}} \exp\left(-\left(\frac{\ln(\sigma_f - \gamma) - \ln(\sigma_m)}{\sqrt{2}\sigma_{sl}}\right)^2\right) \quad (11)$$

CDF of this model can be expressed as:

$$f_{ln} = \frac{1}{2} + \frac{1}{2} \operatorname{erf}\left[\frac{\ln(\sigma_f - \gamma) - \ln(\sigma_m)}{\sqrt{2}\sigma_{sl}}\right] \quad (12)$$

Eq.12 can be rearranged as

$$\sigma_f = \exp\left(\ln(\sigma_m) + \sqrt{2}\sigma_{sl} \operatorname{erf}^{-1}(2f_{ln} - 1)\right) + \gamma \quad (13)$$

Similar to two-parameters Lognormal model, using Arrhenius Eq, Eq 13 can be represented as:

$$\sigma_f = \exp\left(\left\{K_t + \frac{M_t}{T}\right\} + \left\{\sqrt{2}\sigma_{sl} \operatorname{erf}^{-1}(2f_{ln} - 1)\right\}\right) + \gamma \quad (14)$$

Fatigue life prediction model can be presented as

$$\sigma_f = \exp\left(\left\{K_t + L \ln(\epsilon) + \frac{M_t}{T}\right\} + \left\{\sqrt{2}\sigma_{sl} \operatorname{erf}^{-1}(2f_{ln} - 1)\right\}\right) + \gamma \quad (15)$$

1.4 Lognormal Distribution with Power Law

The PDF of this model is given by:

$$F_{ln} = \frac{1}{(\sigma_f)\sigma_{sl}\sqrt{2\pi}} \exp\left(-\left(\frac{\ln(\sigma_f) - \ln(\sigma_m)}{\sqrt{2}\sigma_{sl}}\right)^2\right) \quad (16)$$

CDF of this model can be expressed as:

$$f_{ln} = \frac{1}{2} + \frac{1}{2} \operatorname{erf}\left[\frac{\ln(\sigma_f) - \ln(\sigma_m)}{\sqrt{2}\sigma_{sl}}\right] \quad (17)$$

Eq.17 can be rearranged as

$$\sigma_f = \exp\left(\ln(\sigma_m) + \sqrt{2}\sigma_{sl} \operatorname{erf}^{-1}(2f_{ln} - 1)\right) \quad (18)$$

using Power Law Eq, Eq 18 can be represented as:

$$\sigma_f = \exp\left(\left\{K_t + M_t \ln(T)\right\} + \left\{\sqrt{2}\sigma_{sl} \operatorname{erf}^{-1}(2f_{ln} - 1)\right\}\right) \quad (19)$$

Fatigue life prediction model can be presented as

$$\sigma_f = \exp\left(\left\{K_t + L \ln(\epsilon) + M_t \ln(T)\right\} + \left\{\sqrt{2}\sigma_{sl} \operatorname{erf}^{-1}(2f_{ln} - 1)\right\}\right) \quad (20)$$

1.5 Normal Distribution

The CDF of this model is given by:

$$f_n = \frac{1}{2} + \frac{1}{2} \operatorname{erf}\left(\frac{\sigma_f - \sigma_m}{\sqrt{2}\sigma_{sl}}\right) \quad (21)$$

PDF of this model can be expressed as:

$$F_n = \frac{1}{\sigma_{sl}\sqrt{2\pi}} \exp\left(-\left(\frac{\sigma_f - \sigma_m}{\sqrt{2}\sigma_{sl}}\right)^2\right) \quad (22)$$

Eq.21 can be rearranged as

$$\sigma_f = \sigma_m + \sigma_{sl}\sqrt{2}\operatorname{erf}^{-1}(2f_n - 1) \quad (23)$$

Using Arrhenius Eq in Eq 23:

$$\sigma_f = \left(P_t + \frac{R_t}{T}\right) + \left(\sigma_{sl}\sqrt{2}\operatorname{erf}^{-1}(2f_n - 1)\right) \quad (24)$$

Fatigue life prediction model can be presented as

$$\sigma_f = \left(P_t + Q_t \ln(\epsilon) + \frac{R_t}{T}\right) + \left(\sigma_{sl}\sqrt{2}\operatorname{erf}^{-1}(2f_n - 1)\right) \quad (25)$$

1.6 Gumbell Distribution

The CDF of this model is given by:

$$f_{gb} = \exp\left(-\exp\left(-\frac{\sigma_f - \mu}{\sigma_m}\right)\right) \quad (26)$$

PDF of this model can be expressed as:

$$F_{gb} = \frac{1}{\sigma_m} \left(-\exp\left(-\frac{\sigma_f - \mu}{\sigma_m}\right) + e^{-z}\right) \quad (27)$$

Eq.26 can be rearranged as

$$\sigma_f = -\sigma_m \ln(-\ln(f_{gb})) + \mu \quad (28)$$

Using Arrhenius Eq in Eq 28:

$$\sigma_f = -\sigma_m \ln(-\ln(f_{gb})) + \left(P_t + \frac{R_t}{T}\right) \quad (29)$$

Fatigue life prediction model can be presented as

$$\sigma_f = -\sigma_m \ln(-\ln(f_{gb})) + \left(P_t + Q_t \ln(\epsilon) + \frac{R_t}{T}\right) \quad (30)$$

1.7 Exponential Distribution

The CDF of this model is given by:

$$f_{exp} = 1 - \exp\left(-\frac{\sigma_f}{\sigma_m}\right) \quad (31)$$

PDF of this model can be expressed as:

$$F_{exp} = \frac{1}{\sigma_m} \exp\left(-\frac{\sigma_f}{\sigma_m}\right) \quad (32)$$

Eq.31 can be rearranged as

$$\sigma_f = -\sigma_m \ln(1 - f_{exp}) \quad (33)$$

Using Arrhenius Eq in Eq 33:

$$\sigma_f = -\left(P_t + \frac{R_t}{T}\right) \ln(1 - f_{exp}) \quad (34)$$

Fatigue life prediction model can be presented as

$$\sigma_f = -\left(P_t + Q_t \ln(\epsilon) + \frac{R_t}{T}\right) \ln(1 - f_{exp}) \quad (35)$$

1.8 Gamma Distribution

The CDF of this model is given by:

$$f_{gm}(x) = \frac{1}{\Gamma(m)} \gamma\left(m, \frac{\sigma_f}{\sigma_m}\right) \quad (36)$$

PDF of this model can be expressed as:

$$F_{gm}(x) = \frac{1}{\sigma_m^\alpha \Gamma(m)} \sigma_f^{\alpha-1} \exp\left(-\frac{\sigma_f}{\sigma_m}\right) \quad (37)$$

Eq.36 can't be written in terms of σ_f , we have to do iteration to get σ_f